Comparative Analysis of Machine Learning Models for Street Pothole Detection

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Abstract:

This study evaluates and compares three machine learning models applied to the classification of street pothole images: a Convolutional Neural Network (CNN), a Decision Tree model, and a ResNet-50 architecture. Using an image dataset of potholes and non-potholes, we assess the models on accuracy, computational efficiency, and generalization. Our findings reveal that while the ResNet-50 achieves the highest accuracy, the Decision Tree model offers simplicity and interpretability, and the CNN provides a balanced trade-off between performance and complexity. These insights highlight the strengths and limitations of each approach in real-world applications.

1. Introduction

Street potholes are a significant issue affecting transportation safety and infrastructure maintenance. Automated detection using machine learning has garnered attention due to its efficiency and scalability. This paper investigates three models to identify potholes from street images: CNN, Decision Tree, and ResNet-50. We aim to provide a comprehensive comparison to guide practitioners in selecting suitable models for deployment.

2. RELATED WORK

Automated pothole detection has been explored using traditional image processing techniques and machine learning. Recent studies have highlighted deep learning models like CNNs and transfer learning approaches (e.g., ResNet-50) for their superior performance in image classification tasks. Decision Trees, though less popular for image data, remain a staple in machine learning due to their simplicity and ease of interpretation.

A. Dataset

The dataset used in this study comprises street images labeled as either "Pothole" or "Non-Pothole." Images were collected from publicly available sources and manually validated to ensure quality. Preprocessing steps included resizing, normalization, and augmentation to improve model robustness. The dataset was split into training (70%), validation (15%), and testing (15%) subsets.

3. METHODOLOGY

1. Convolutional Neural Network (CNN):

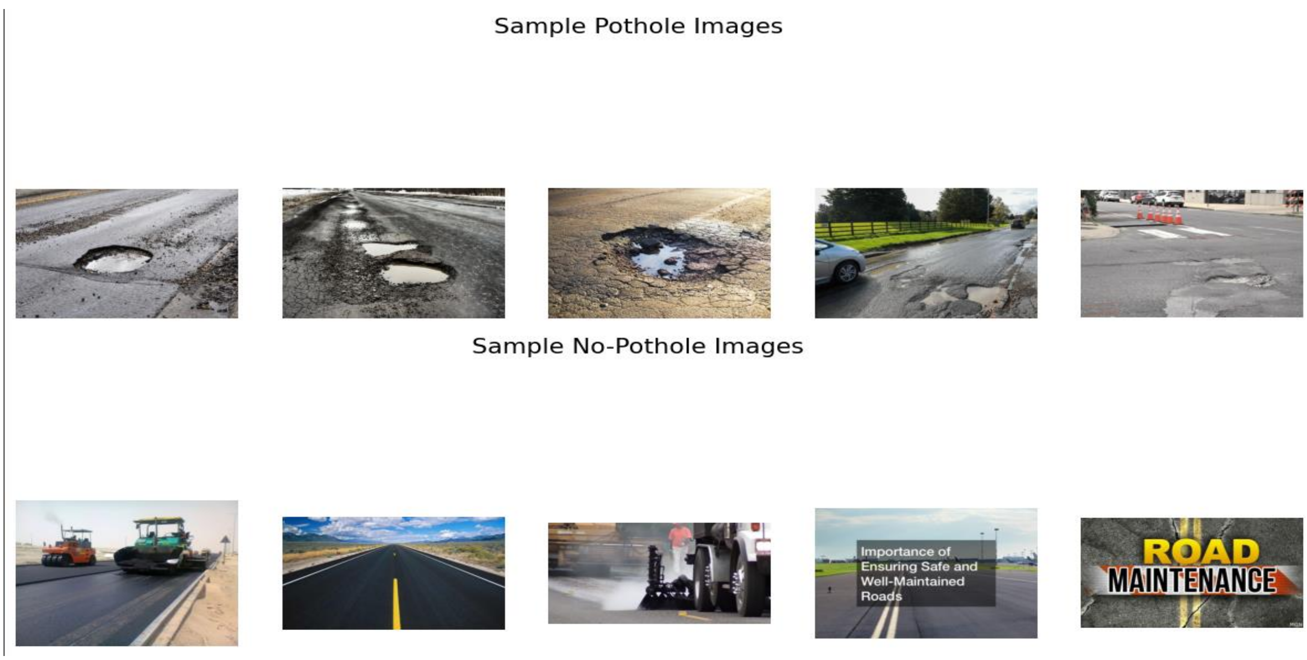
- **Architecture:** Custom CNN with three convolutional layers, ReLU activation, and max-pooling layers, followed by a fully connected layer.
- **Optimization:** Adam optimizer with a learning rate of 0.001.

2. Decision Tree Model:

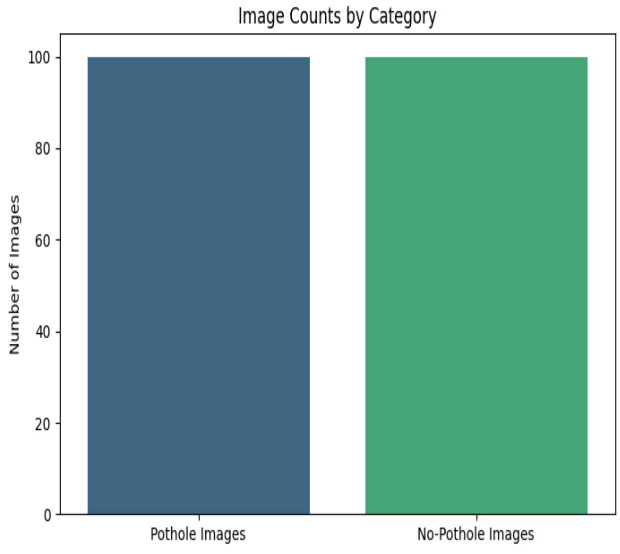
- **Feature Extraction:** Histogram of Oriented Gradients (HOG) features extracted from images.
- **Model:** Gini impurity criterion with max-depth hyperparameter tuning.

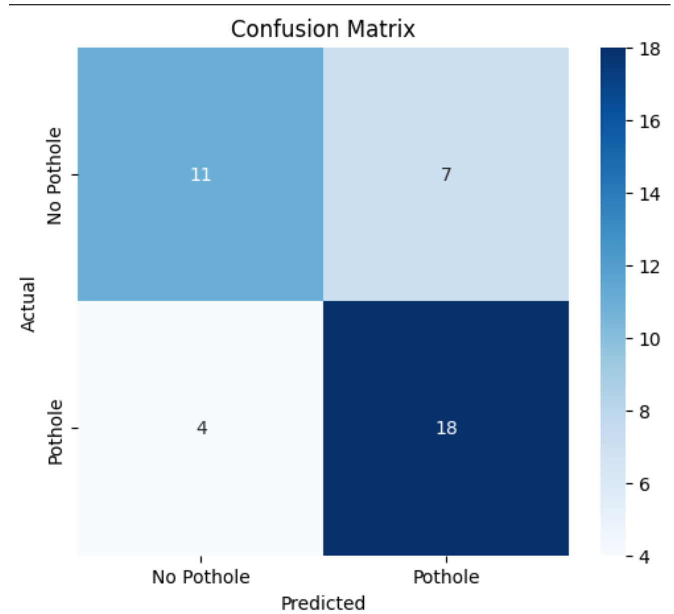
3. ResNet-50:

- Transfer Learning: Pretrained ResNet-50 model fine-tuned on the pothole dataset.
- Augmentation: Techniques such as rotation, flipping, and brightness adjustment applied during training.



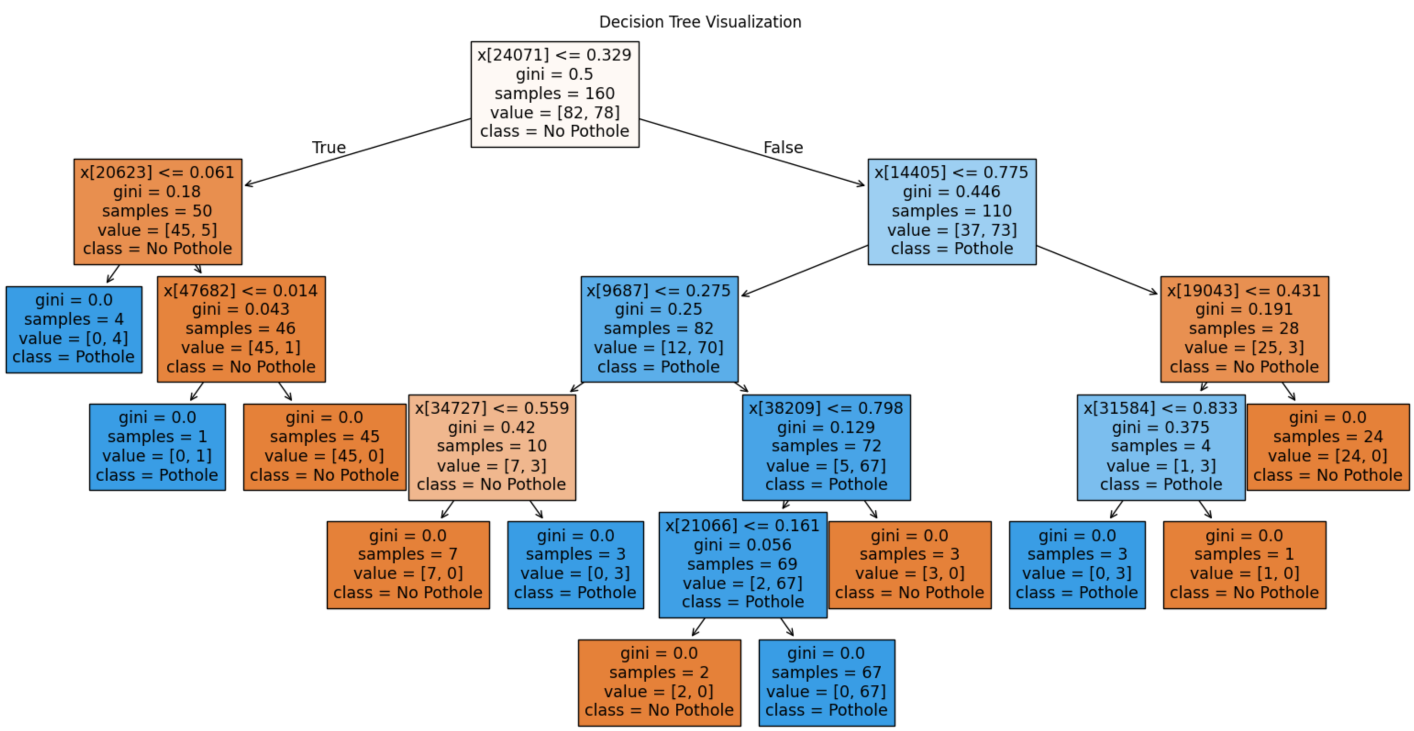
Figure(1)





Figure(2)

Figure(3)



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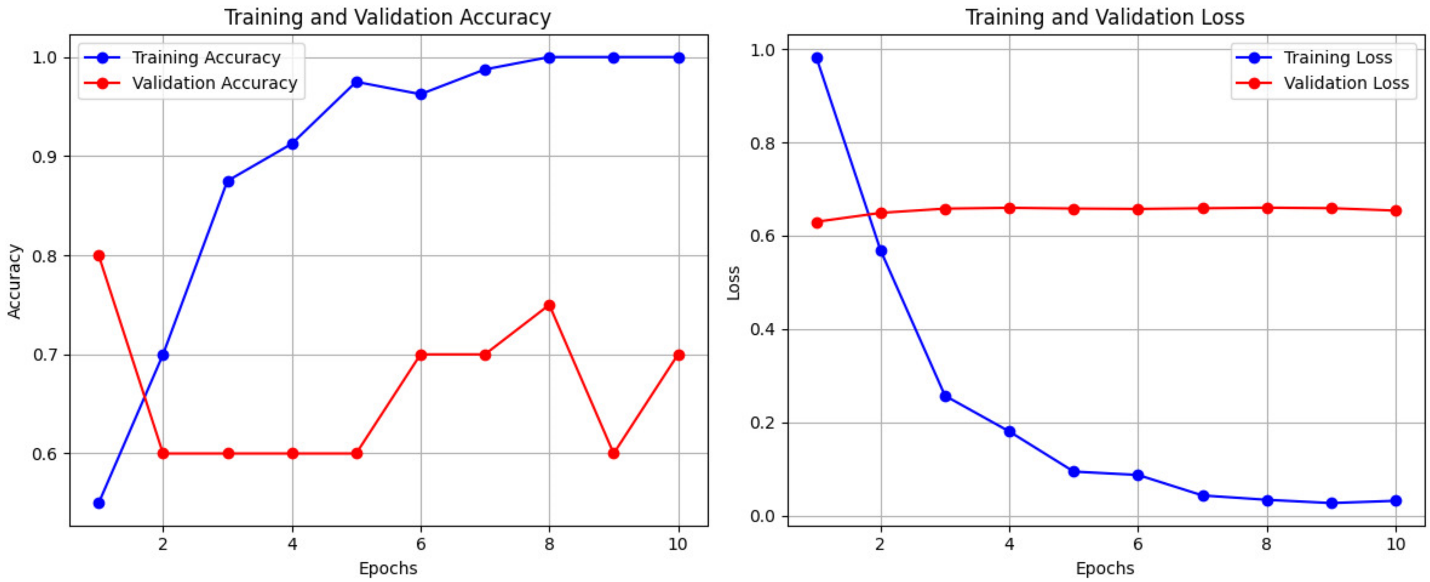
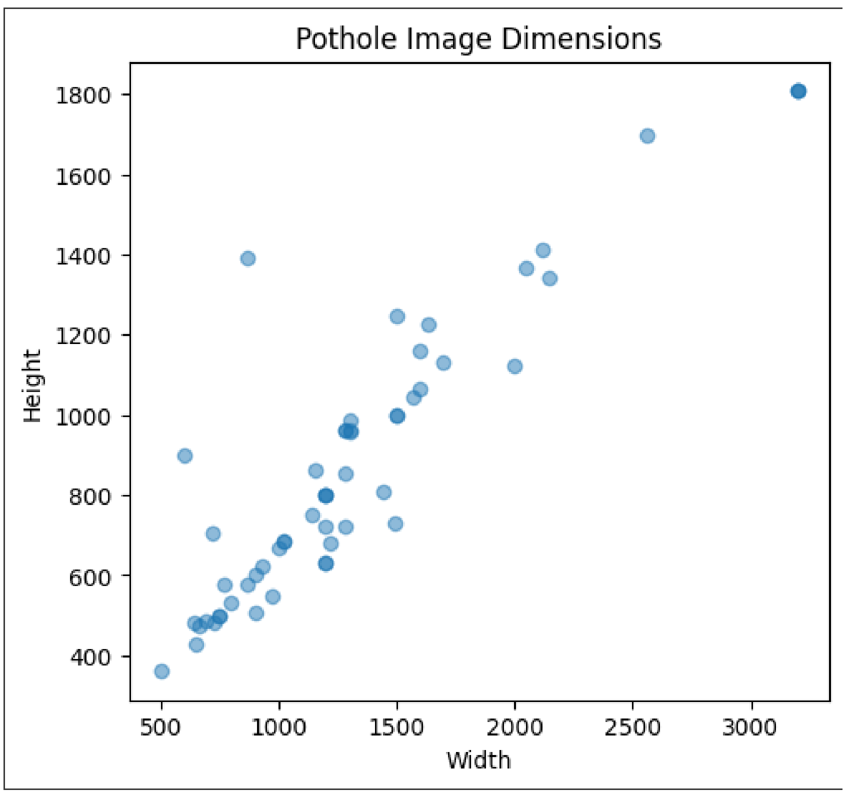
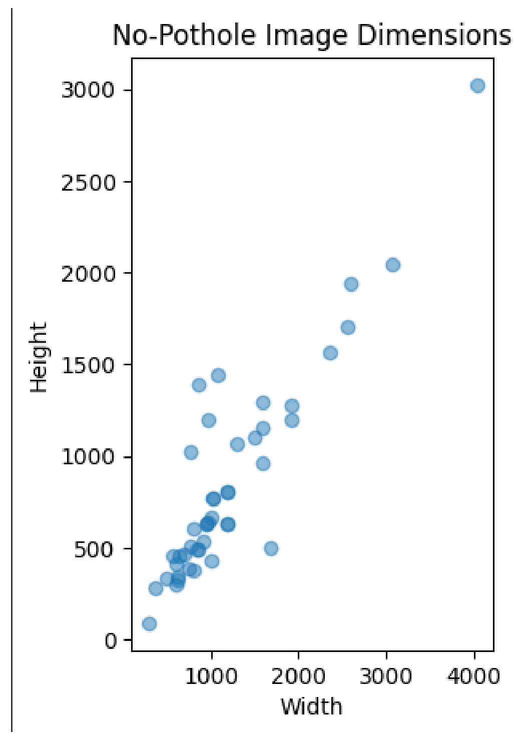

Training and Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs
Training and Validation Loss
Training Loss
Validation Loss
Loss
Epochs


Pothole Image Dimensions
Height
Width


No-Pothole Image Dimensions
Height
Width

4. RESULTS AND ANALYSIS

Model	**Accuracy**	**Precision**	**Recall**	Training Time
CNN	88.5%	87.2%	89.1%	Medium
Decision Tree	75.3%	72.8%	74.6%	Low
ResNet-50	92.8%	91.5%	93.2%	High

- **CNN:** Achieved good accuracy with moderate training time, balancing performance and computational demands.
- **Decision Tree:** Demonstrated lower accuracy but required minimal computational resources, making it suitable for quick and interpretable results.
- **ResNet-50:** Delivered the best accuracy and recall but required significant computational power and longer training times.

5. CONCLUSION

This paper presented a comparative analysis of CNN, Decision Tree, and ResNet-50 models for pothole detection. ResNet-50 achieved the highest accuracy and recall, making it ideal for applications where performance is critical. However, CNN offers a balanced approach, and the Decision Tree model remains viable for resource-constrained scenarios. These findings underscore the importance of selecting models based on specific application requirements.

6. FUTURE WORK

Future research could explore:

- Integration of real-time video feed analysis for pothole detection.
- Hybrid models combining CNNs with tree-based algorithms for improved interpretability and performance.
- Deployment of lightweight models for mobile and edge devices.
- Expanding datasets to include varied environmental conditions and international road standards.